

Zewail City of Science, Technology and Innovation

University of Science and Technology

School of Computational Sciences and Artificial Intelligence

GRADUATION PROJECT PROPOSAL

Fekry - Mobile Personal Assistant

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1. Project Title

Fekry - An Android Personal Assistant for Egyptian Arabic and Arabizi Speakers

2. Team Members

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3. Supervisors

Role	Name	Department / Affiliation	Email
Main Supervisor	Dr. Mayada Mansour Ali Hadhoud	DSAI/Associate Professor	mhadhoud@zewailcity.edu.eg
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4. Project Classification

Item	Your Project
Research Area(s)	Artificial Intelligence, Natural Language Processing, Speech Processing and Software Engineering
Research Focus	Context-aware understanding of Egyptian Arabic, English, code-switching and typed Arabizi; speech/text inputs; intent and entity extraction; authorized multi-step task execution.
Applied Field	Personal productivity, educational/study planning, and voice-accessible Android assistance.
Potential Industrial Partner	NONE

5. Abstract

FEKRY is a proposed AI-powered personal assistant delivered as an Android mobile application. It aims to help Egyptian Arabic-speaking users carry out selected daily and academic tasks through natural conversation, with voice as a central interaction method. The application will support everyday Egyptian Arabic, English, mixed Arabic-English requests, and typed Arabizi. Text and basic image or screenshot input will complement spoken requests.

The project focuses on understanding a request in its conversational context, identifying the required tasks, asking for missing information, and carrying out supported steps with the user's approval. For

example, a user can describe an upcoming exam, provide the topics and available study time, and ask FEKRY to propose sessions and reminders. Follow-up requests can refer to the plan already being discussed without requiring the user to repeat all its details.

Personalized assistance will cover task and reminder management, calendar-based planning, and email preparation. Study planning will use information supplied by the user together with permitted calendar data. Preferences and selected memories will remain visible and controllable by the user. External actions, such as creating calendar events or sending an email, will require the relevant account permissions and explicit confirmation of the action details.

The mobile application will connect to backend AI services and selected official application APIs. Models and implementation methods will be selected through experiments against the required functions, available data, computational resources, and licensing conditions. The expected outcome is an integrated prototype evaluated for request understanding, context handling, speech quality, task completion, security, privacy, response time, and usability. Its contribution will be assessed on defined Egyptian Arabic and mixed-language scenarios.

6. Problem Statement

Users may express a goal through several conversational turns, refer to earlier information, mix Egyptian Arabic with English, or omit details that they expect an assistant to ask about. Translating such communication into the correct tasks, dates, recipients, and application actions is the practical problem addressed by FEKRY. Users who prefer voice interaction also need a clear way to review a proposed action and understand whether it was completed.

Existing assistants already support capabilities beyond simple questions and isolated commands. Official documentation describes connected productivity workflows in Gemini, contextual and application-related assistance in Siri, and voice tasks and combined commands in Bixby [1-3]. FEKRY therefore does not assume that these capabilities are absent from the market. It proposes a focused implementation and evaluation of selected personal-assistance workflows for everyday Egyptian Arabic, English, typed Arabizi, and mixed requests.

Egyptian Arabic written in Arabic or Latin script presents practical language-understanding challenges. Research such as Nile-Chat and resources such as BOLT provide relevant background for dual-script Egyptian text [4, 5]. FEKRY will address the forms used in its selected scenarios; identifying regional dialects or classifying speakers by governorate is outside the project scope.

The motivation is to connect natural communication to useful, controlled actions in a manageable mobile prototype. The system should retain relevant conversation context, distinguish a suggestion from an executed action, clarify ambiguous information, and use personal data only within the permissions granted by the user.

7. Motivation

Egyptian Arabic and its Latin-script writing form create practical challenges for normalization and conversational understanding; code-switching further complicates speech and text processing. FEKRY connects this language work to useful productivity actions rather than isolated command classification. Its focused study-planning, reminders and email workflows can support everyday organization while allowing voice-first interaction. The technical contribution is a reproducible, consent-based evaluation of language and speech components, contextual task orchestration and permission-aware execution. User review, editable memory and explicit consent make trustworthy behavior part of the solution, not an optional extension.

8. Objectives

1. Deliver a working Android app and backend demonstrating the agreed functions, including multi-turn corrections, confirmation screens, and Google Calendar/Gmail integrations.
2. Evaluate Egyptian Arabic, English, typed Arabizi and mixed-language requests on a held-out benchmark, targeting at least 85% intent Macro-F1 and reporting date/time and other field extraction separately.
3. Assess foreground wake-word detection (at least 90% recall; no more than one false activation per hour) and speech recognition (WER no worse than 30% in clean test conditions).
4. Measure typed Arabizi normalization (CER no worse than 0.20) and achieve at least 80% completion of defined contextual workflows, including clarifications and chained actions.
5. Implement permission checks, action previews and confirmations; target zero unauthorized external actions and zero cross-user data access in the specified security test suite.
6. Report latency (typical text response about 5 s; voice response about 10 s, with median/P95), held-out model comparisons, error and robustness analysis, and majority-positive feedback from a consented user pilot.

9. Project Scope

Included (In Scope)	Excluded (Out of Scope)
F1: Always-on/background wake-word service.Create, view, edit and complete FEKRY tasks and reminders; clarify ambiguous dates/times.	Regional dialect or governorate classification; complete coverage of all Arabic dialects.
F2: Read authorized Google Calendar availability; propose, revise and confirm study sessions/events from user-provided constraints.	Automatic discovery of all academic requirements; unsupported services or unapproved actions.
F3: Draft/revise Gmail messages; preview recipient, subject and body; send only after consent and confirmation.	Unrestricted access to installed apps, privileged device functions or permission bypasses.
F4: Retain conversational state, resolve follow-up references, split compound tasks and report partial failures.	Complete iOS product, general computer automation, or a large smart-home ecosystem.
F5: Let users review, edit, disable and delete selected preferences/memory and disconnect external accounts.	Speaker identification, production voice cloning, emotional-state analysis and broad visual reasoning.
F6:Voice/push-to-talk; foreground wake-word trials; English/Egyptian Arabic/mixed speech; Arabizi text; OCR of user-selected task-relevant images.	unrestricted camera inference, or automated device control outside Android/API limits.

10. Expected Outcomes

Deliverable / Outcome	Description	Target Metric
Integrated Android assistant	Runnable mobile/backend system, persistent tasks/preferences and F1–F5 demonstrations with official Google integrations.	All five workflows demonstrated by Week 12 in spring.
Language understanding	Intent and required-detail recognition for Egyptian Arabic, English, mixed requests and Arabizi.	Intent Macro-F1 ≥ 85%; report field extraction.
Speech and wake word	Speech-to-text, spoken responses, push-to-talk and foreground wake-word activation.	WER ≤ 30% (clean); recall ≥ 90%; ≤ 1 false wake/h.
Arabizi normalization	Human-reviewed Latin-script Egyptian examples and normalization/error analysis.	Character Error Rate ≤ 0.20.
Contextual actions	Multi-turn clarifications, corrections, scheduling revisions and chained F1–F5 actions.	Workflow completion ≥ 80% on defined tests.
Security and privacy	Authorized account data, editable/deletable memory, action previews and explicit confirmations.	0 unauthorized actions; 0 cross-user leaks in tests.
Response time and UX	End-to-end mobile-to-service tests, spoken/text responses and consenting pilot.	Typical text ≈ 5 s; voice ≈ 10 s; majority positive feedback.
Benchmark and experiments	A prepared test dataset of FEKRY commands, conversations, and wake-word recordings, used to evaluate system performance and compare different models. The evaluation will include baseline comparisons, experiments to understand the effect of individual components, and analysis of errors.	Documented evaluation results, model comparisons, and error analysis using held-out test data and separate speakers for speech testing.
Final academic deliverables	Source repository, GP1/GP2 reports, installation/user guides, final poster and 3–4-minute video.	GP2 final defense and deliverables per official timeline.

Data and validation: Candidate resources include BOLT Egyptian Arabic SMS/Chat [5] (LDC license), Nile-Chat [4], ArzEn-MultiGenre v4 [6] (parallel text), Hamed et al. [7] (code-switched speech) and user-consented FEKRY commands/speech. External access and licenses must be verified; private Gmail/calendar content will not be used for training by default. Separate speaker-disjoint speech splits, session separation, de-duplicated text splits, equivalent-model comparisons, bias/robustness checks, pilot feedback and NIST/OWASP-informed security tests will be documented [14], [15].

11. Preliminary Literature Review

The following entries retain the proposal's reference numbers. They cover established assistants, relevant language/speech data, and candidate model families; none is a measured FEKRY benchmark.

11.1 Summary of Related Work

Ref.	Author(s), Year	Method / Approach	Key Findings	Limitations / Gap
[1]	Shang et al., 2025	Nile-Chat Egyptian Arabic and Latin-script models.	Relevant dual-script Egyptian language resources.	No guarantee of task-specific F1–F5 behavior.
[2]	Chen et al., 2017	BOLT Egyptian Arabic SMS/chat and transliteration.	Useful candidate data for normalization and understanding.	Licensed LDC data; access not guaranteed.
[3]	Al-Sabbagh, 2024	ArzEn-MultiGenre v4 parallel Egyptian/English text.	Relevant Egyptian Arabic–English parallel examples.	Licensed LDC data; access not guaranteed.
[4]	Hamed et al., 2018	Code-switched Egyptian Arabic–English speech corpus.	Relevant mixed-language speech study resource.	Access/reuse permissions must be checked.
[5]	Yang et al., 2025	Qwen3 pretrained language-model family.	Candidate generation/reasoning backbone.	Needs accuracy, latency and license evaluation.
[6]	Dang et al., 2024	Aya Expanse multilingual generative models.	Candidate multilingual language generation.	Not validated for FEKRY-specific workflow safety.
[7]	Abdul-Mageed et al., 2021	ARBERT/MARBERT bidirectional Arabic encoders.	Candidate adapted intent/classification baseline.	Encoder, not a complete agent or action system.

11.2 Research Gap

Existing products show that connected workflows, voice interaction and contextual help are established capabilities [1]–[3]. The specific research opportunity is not to assume those systems lack Egyptian Arabic support, but to prepare a transparent benchmark and measure selected end-to-end Egyptian Arabic, mixed-language and typed-Arabizi productivity workflows. The academic resources above support parts of the problem—dual-script text, parallel data, code-switching speech or pretrained models—but are not by themselves a validated assistant dataset or secure application. FEKRY will compare suitable baselines on held-out examples, evaluate context and field extraction, trace failures across speech/understanding/planning/execution, and require explicit review before actions. Claims will be limited to measured scenarios, participant groups, and permissions.

12. Timeline

#	Phase	Key Activities	Start	End	Milestone
1	GP1: Team formation	Confirm title, roles, supervisor; create/share Git repository.	Fall W 1	Fall W1	Team and Git repo.
2	GP1: Proposal	Complete abstract, problem, scope, outcomes, plan and AI-use statement.	Fall W4	Fall W4	Project proposal
3	GP1: Market & literature	Compare assistants, document language research and gaps.	Fall W4	Fall W6	Literature/market report
4	GP1: Mid-semester	Present background, early progress and next steps.	Fall W6	Fall W8	15-minute presentation

#	Phase	Key Activities	Start	End	Milestone
5	GP1: Requirements	SRS, F1–F5 use cases; data sources, labels/splits; privacy/methods.	Fall W8	Fall W10	SRS and research plan
6	GP1: Design & PoC	Architecture, ERD, UML, Gantt, division of work, feasibility baseline.	Fall W10	Fall W12	Design + proof of concept
7	GP1: Draft report	Write GP1 draft and capture supervisor revisions.	Fall W12	Fall W13	GP1 final defense
8	GP1: Final report	Finalize research/design report and prepare Q&A.	Fall W13	Fall W15	Working version + report
9	GP2: Implementation	Mobile/backend implementation, DB, basic tests and progress report.	Spring W1	Spring W4	MVP demonstratio n.
10	GP2: MVP	Mobile/backend implementation, DB, basic tests and progress report.	Spring W4	Spring W9	15-minute progress presentation.
11	GP2: Midterm	Review integration, MVP evidence and outstanding issues.	Spring W9	Spring W10	Progress Report 2 + metrics.
12	GP2: Evaluation	Held-out tests, model comparisons, ablations, usability/security checks.	Spring W9	Spring W11	GitHub + live F1–F5 demo.
13	GP2: Final software	Complete runnable Android/backend, source and integration demo.	Spring W11	Spring W12	Final documentatio n.
14	GP2: Documentation	Write final thesis, compliance, guides, references and presentation.	Spring W12	Spring W14	Poster/video submission.
15	GP2: Poster & video	Show approach, features and measured results in poster and 3–4 min video.	Spring W12	Spring W14	Final defense.
16	GP2: Final defense	Demonstrate product, documentation, poster and video; individual Q&A.	After finals	After finals	

References

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[14] National Institute of Standards and Technology, “AI Risk Management Framework.” https://www.nist.gov/itl/ai-risk-management-framework

[15] OWASP GenAI Security Project, “OWASP Top 10 for LLM Applications.” https://genai.owasp.org/llm-top-10/